Programming paradigms: procedural and object-oriented

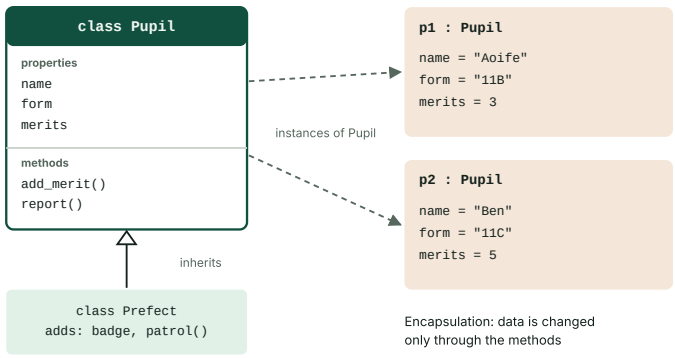
D01 FACT FILE

Specification: students should be able to

- Describe the following programming paradigms: procedural programming and object-oriented programming

CCEA fact file: Software Development Trends

Key words: paradigm, procedural, imperative, sequential, procedure, function, object-oriented, object, class, property, method, inheritance, encapsulation



A Pupil class, two objects created from it, and a Prefect class that inherits from it

Procedural programming

In procedural programming the programmer gives the computer a sequence of instructions to follow, one after another. The language is imperative, meaning each statement is a command. Larger programs are broken down (top-down design) into procedures and functions, which are named blocks of code that carry out one task and can be called from anywhere in the program. Data is usually held in variables that are passed into these procedures.

Object-oriented programming

In object-oriented programming the solution is designed around objects. A class is a template describing what every object of that type will know (its properties or attributes) and do (its methods). An object is one instance created from the class, with its own values. Objects communicate by calling each other's methods. Encapsulation keeps an object's data inside it so it can only be changed through its methods. Inheritance lets a new class reuse and extend an existing class. Because classes are self-contained they are easier to test and can be reused in other programs.

T Key terms from the mark schemes

Procedural programming
Step-by-step instructions executed in sequence, organised into procedures and functions.
Credited: U4 2019 Q7, 2024 Q5, 2025 Q7

Object-oriented programming
Programs built from self-contained objects that combine data (properties) and methods.
Credited: U4 2024 Q5, 2025 Q7

Class
A blueprint defining the properties and methods of objects.
Credited: U4 2025 Q7

Object
An instance created from a class with its own data.
Credited: U4 2025 Q7

Encapsulation
Keeping an object's data inside it, accessed only through its methods.
Credited: U4 2025 Q7

Inheritance
A class taking on the properties and methods of another class.
Credited: U4 2025 Q7

C C# example

Procedural programming

```
static double Area(double width, double height)
{
    return width * height;
}

Console.Write("Width: ");
double w = Convert.ToDouble(Console.ReadLine());
```

```
Console.Write("Height: ");
double h = Convert.ToDouble(Console.ReadLine());
Console.WriteLine("Area is " + Area(w, h));
```

Exam focus

Expect fill-the-gap questions on paradigm vocabulary (2024 Q5(a)) and a QWC description of both approaches (2025 Q7). Write in paragraphs, cover both paradigms in similar depth, and use the technical terms – generic answers stay in Level 1.

Unit 4 2025 Q7	[6]	Procedural vs object-oriented programming (QWC)
Unit 4 2024 Q5	[9]	Paradigms fill-in; compilers and interpreters true/false

? Check yourself

1. Describe procedural programming. [3]
2. State the difference between a class and an object. [2]
3. In OOP, what is meant by a property and a method? [2]

Answers: 1. Any three from: the programmer writes step-by-step instructions telling the computer exactly what to do (1) / it is imperative – each statement is a command (1) / instructions are executed sequentially, one after another, so the order is the program flow (1) / code is grouped into procedures and functions that can be called when needed (1) / the program is designed top-down – the main task is broken into smaller sub-tasks (1) / data is held in variables, kept separate from the code, and passed into procedures (1) 2. A class is a blueprint/template that defines the properties and methods for a type of object (1) / an object is a single instance created from a class, with its own data values (1). Accept e.g. class Pupil; objects p1 and p2. 3. A property (attribute) is an item of data that an object holds, e.g. a Pupil's Name or Form (1) / a method is an action/subroutine an object can carry out, e.g. AddMerit() (1)

Development environments, editing features and translators

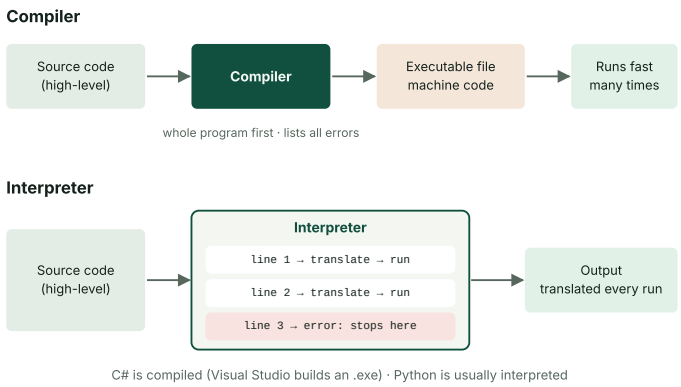
D02 FACT FILE

Specification: students should be able to

- Explain the significance of the following aspects of software development environments: editing features; high-level code translation and execution

CCEA fact file: Software Development Trends

Key words: source code, machine code, high-level language, IDE, code editor, IntelliSense, auto-complete, code outlining, line numbering, syntax highlighting, debugger, translator, compiler, interpreter



A compiler translates the whole program into an executable; an interpreter translates and runs one line at a time

Why translation is needed

High-level languages let programmers concentrate on solving the problem rather than on how the processor works. The processor, however, only executes machine code, so every high-level program must be translated. A compiler translates the whole program into an executable file before it runs; an interpreter translates and runs it one statement at a time.

Integrated development environments

An IDE combines a code editor, translator, debugger, run-time environment and often a GUI designer in a single package. Editing features such as IntelliSense, auto-complete, syntax highlighting, error underlining, line numbers and code outlining help programmers write code faster and with fewer errors. Debugging tools such as breakpoints and stepping let the programmer watch variables change, which helps find logic errors.

T Key terms from the mark schemes

IDE
Integrated development environment: editor, translator, debugger and tools in one application.
Credited: U4 2019 Q1, 2022 Q3, 2025 Q3

Translator
Software that converts source code into machine code: a compiler or interpreter.
Credited: U4 2019 Q1

Compiler
Translates the whole program into machine code before running, producing a separate executable.
Credited: U4 2022 Q3, 2024 Q5

IntelliSense / auto-complete
Editor feature that suggests or completes code as it is typed.
Credited: U4 2019 Q1, 2025 Q3

Machine code
Binary instructions the processor executes directly.
Credited: U4 2019 Q1, 2022 Q3

Interpreter
Translates and executes source code one line at a time.
Credited: U4 2024 Q5

Exam focus

You may be asked to name editor features (2019 Q1(d), 2025 Q3(c)(i)), complete a paragraph on translation (2022 Q3(a)), judge compiler/interpreter statements (2024 Q5(b)) or write a QWC answer on development environments (2023 Q8). Always say how the feature helps the programmer.

Unit 4 2025 Q3	[15]	Musician hire algorithm; variables and data types; IDE aids; syntax vs logic
Unit 4 2024 Q5	[9]	Paradigms fill-in; compilers and interpreters true/false
Unit 4 2023 Q8	[6]	Editing features, translation and execution (QWC)
Unit 4 2022 Q3	[6]	Translating code; IDE features
Unit 4 2019 Q1	[18]	Data types, range and presence checks, extending a program, IDE features, translation

? Check yourself

1. Explain why source code written in a high-level language must be translated. [2]
2. Describe two editing features of an IDE and how each helps a programmer. [4]
3. Describe how a compiler translates a program. [2]

Answers: 1. The CPU can only execute machine code (binary instructions) (1) / so high-level source code, which is close to English, must be converted into machine code by a translator (compiler or interpreter) before it can run (1) 2. Any two features, each named (1) + how it helps (1), e.g.: IntelliSense/auto-complete (1) – suggests keywords, variable and function names and shows parameters as you type, so fewer typing and syntax errors and faster coding (1) / syntax highlighting/colour coding (1) – keywords, strings and comments in different colours so code is easier to read and mistakes stand out (1) / syntax error highlighting (wavy underline) (1) – errors are spotted immediately as you type (1) / line numbering (1) – find a line quickly when an error message gives a line number (1) / code outlining (1) – collapse/expand blocks so long programs are easier to navigate (1) / auto-indent (1) – keeps indentation consistent so structure is clear (1) 3. The compiler translates the whole source program into machine code in one go before it runs (1) / producing a separate executable file; it reports all errors as a list after compiling and will not produce the executable until syntax errors are fixed (1)

Binary, units of data and denary–binary conversion

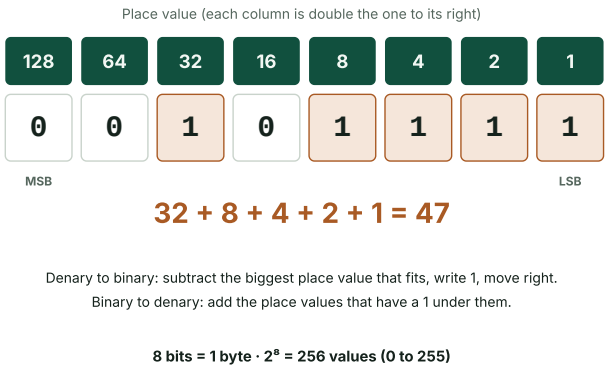
D03 FACT FILE

Specification: students should be able to

- Describe how a number is converted to a binary pattern for storage in a computer
- Demonstrate understanding of the following units of data: bit, nibble, byte, kilobyte, megabyte, gigabyte and terabyte
- Demonstrate understanding of and use number representation and convert between denary, binary and hexadecimal (denary–binary)

CCEA fact file: Digital Data

Key words: binary, denary, bit, nibble, byte, place value, kilobyte, megabyte, gigabyte, terabyte, most significant bit, least significant bit



Place values of an 8-bit byte showing 47 as 00101111

Binary patterns

A computer stores every number as a pattern of bits. Each column in binary is worth twice the column to its right: 128, 64, 32, 16, 8, 4, 2, 1 for one byte. To convert denary to binary either subtract the largest place value that fits and move right, or divide by 2 repeatedly and read the remainders upwards. To convert binary to denary, add the place values of every column that holds a 1. Always show working and write a full 8 bits.

Units

A bit is a single 0 or 1; a nibble is 4 bits; a byte is 8 bits and normally stores one character. Each larger unit is 1024 of the one below: kilobyte, megabyte, gigabyte, terabyte. With n bits there are 2^n possible patterns, so one byte can hold 256 values from 0 to 255.

T Key terms from the mark schemes

Denary
Base-10 number system.
Credited: U4 2025 Q4

Binary
Base-2 number system using 0 and 1.
Credited: U4 2022 Q6, 2025 Q4

Exam focus

Every Unit 4 paper has a conversion question: 2022 Q6(a) (54), 2023 Q7(b) (47), 2024 Q4(b)–(c) (80 and 10101010), 2025 Q4(a). Unit-matching questions on bit, nibble and byte appeared in 2024 Q4(a) and 2025 Q4(c). Marks are given for working.

Unit 4 2025 Q4	[15]	Binary conversion, addition and overflow, units, Unicode vs ASCII
Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q7	[13]	Bit, 47 to binary, 66 to hex, advantage of hex, binary addition and overflow
Unit 4 2022 Q6	[14]	Binary digit, 54 to binary, nibble, hex, overflow, storage capacity
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Explain what is meant by a bit. [2]
2. Convert the denary number 54 into an 8-bit binary number. Show your working. [2]
3. Convert the binary number 10101010 into denary. Show your working. [2]

Answers: 1. A bit is a binary digit (1) / the smallest unit of data, with a value of either 0 or 1 (off/on) (1) 2. Working, e.g. 32 + 16 + 4 + 2 = 54 under place values 128 64 32 16 8 4 2 1 (or repeated division by 2 with remainders) (1) / 00110110 (1) 3. Working: 128 + 32 + 8 + 2 (1) / = 170 (1)

Hexadecimal

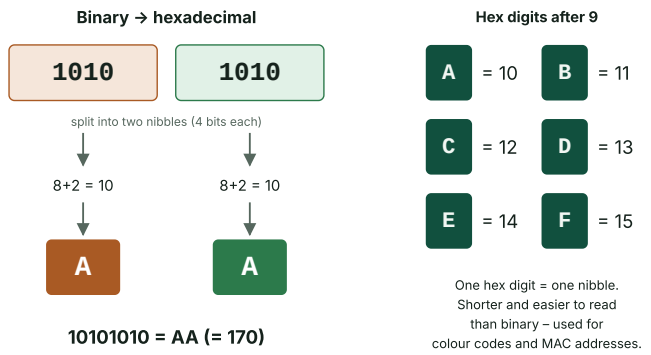
D04 FACT FILE

Specification: students should be able to

- Demonstrate understanding of and use number representation and convert between denary, binary and hexadecimal

CCEA fact file: Digital Data

Key words: hexadecimal, base 16, nibble, hex digit, colour code, MAC address



Converting the byte 10101010 to hexadecimal AA by splitting it into nibbles

Hexadecimal

Hex is base 16, using the digits 0–9 and the letters A–F for 10–15. Because 16 = 2^4, one hex digit represents exactly one nibble, so any byte can be written as two hex digits. Programmers use hex because it is much shorter than binary, easier to read and less error-prone, and it converts directly to binary. Common uses include colour codes, MAC addresses and memory addresses.

Conversions

Binary to hex: split into nibbles and convert each to a hex digit. Hex to binary: write each digit as 4 bits. Denary to hex: divide by 16 – the quotient is the first digit and the remainder the second (or go via binary). Hex to denary: multiply the first digit by 16 and add the second.

T Key terms from the mark schemes

Hexadecimal
Base-16 number system (0–9, A–F); a shorthand for binary.
Credited: U4 2019 Q8, 2023 Q7

Exam focus

Hex conversion appeared in 2019 Q8(b)–(c) (including uses of hex), 2022 Q6(b)(ii), 2023 Q7(c) (66 to hex and an advantage of hex) and 2024 Q4(b)(ii) and (c)(ii). Show the nibble split for working marks.

Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q7	[13]	Bit, 47 to binary, 66 to hex, advantage of hex, binary addition and overflow
Unit 4 2022 Q6	[14]	Binary digit, 54 to binary, nibble, hex, overflow, storage capacity
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Convert the binary number 10101010 into hexadecimal. Show your working. [2]
2. Convert the denary number 66 into hexadecimal. Show your working. [3]
3. Convert the hexadecimal number 3B into denary. [2]

Answers: 1. Split into nibbles: 1010 | 1010 = 10 | 10 (1) / AA (1) 2. Either method: 66 ÷ 16 = 4 remainder 2 (1) / the quotient 4 is the first digit and the remainder 2 the second (1) / 42 (1). Accept via binary: 66 = 01000010 (1) → 0100 | 0010 (1) → 42 (1). 3. 3 × 16 + 11 (B = 11) = 48 + 11 (1) / = 59 (1)

Binary addition and overflow

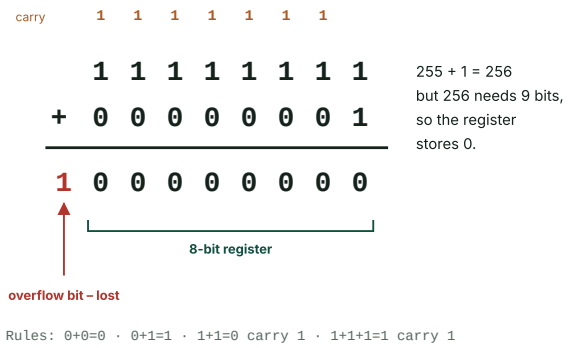
D05 FACT FILE

Specification: students should be able to

- Perform the addition of two bytes and explain the meaning of overflow

CCEA fact file: Digital Data

Key words: binary addition, carry, overflow, register, 8-bit



Adding 11111111 and 00000001 in 8 bits causes overflow

Binary addition

Add from the right-hand column. 0+0 = 0, 0+1 = 1, 1+1 = 0 carry 1, and 1+1+1 = 1 carry 1. Write the carries clearly and check by converting the two numbers and the answer to denary.

Overflow

Overflow happens when the result of a calculation needs more bits than are available. In an 8-bit register the largest value is 255, so any total above 255 produces a carry out of the 128 column that cannot be stored. The extra bit is lost and the stored result is wrong (e.g. 255 + 1 is stored as 0). This can make a program give incorrect results or fail.

Key terms from the mark schemes

Carry
The digit passed to the next column when a binary column total exceeds 1.
Credited: U4 2025 Q4

Overflow
When a result is too large for the available bits, so the extra bit is lost and the result is wrong.
Credited: U4 2019 Q8, 2022 Q6, 2024 Q4, 2025 Q4

Exam focus

Binary addition with overflow appears every year: 2019 Q8(d), 2022 Q6(c), 2023 Q7(d), 2024 Q4(d), 2025 Q4(b). Typical marks: 3 for the addition with carries and 1–2 for explaining overflow and its effect.

Unit 4 2025 Q4	[15]	Binary conversion, addition and overflow, units, Unicode vs ASCII
Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q7	[13]	Bit, 47 to binary, 66 to hex, advantage of hex, binary addition and overflow
Unit 4 2022 Q6	[14]	Binary digit, 54 to binary, nibble, hex, overflow, storage capacity
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Add the binary numbers 00110101 and 00011010. Show the carries. [3]
2. Add 10110110 and 01100101. Show your working. [3]
3. Explain the meaning of overflow. [2]

Answers: 1. 00110101
+ 00011010
= 01001111
Carries shown into the 32 and 64 columns (1) / right-hand bits 1111 correct (1) / final answer 01001111 (1). Check: 53 + 26 = 79. 2. 10110110
+ 01100101
= 1 00011011
Carries shown (carries out of the 4, 32, 64 and 128 columns) (1) / 8-bit result 00011011 (1) / the carry out of the 128 column is a 9th bit – overflow has occurred (1). Check: 182 + 101 = 283, which is > 255; the stored 8 bits are 27. Accept the 9-bit answer 100011011 with overflow identified. 3. Overflow occurs when the result of a calculation is too large (1) / to be stored in the number of bits available, e.g. greater than 255 in an 8-bit register (1)

Character sets: ASCII and Unicode

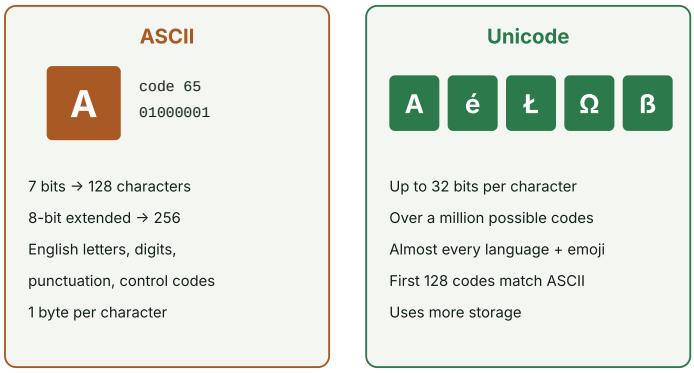
D06 FACT FILE

Specification: students should be able to

- Demonstrate understanding of the following types of character representation: ASCII (7-bit and 8-bit) and Unicode

CCEA fact file: Digital Data

Key words: character set, ASCII, extended ASCII, Unicode, character code, control character, emoji



ASCII compared with Unicode

ASCII

The American Standard Code for Information Interchange originally used 7 bits, giving 128 codes: 32 control codes followed by the space, digits, punctuation and upper and lower case English letters. Extended ASCII uses 8 bits for 256 characters, adding some accented letters and symbols. ASCII is efficient but cannot represent most of the world's languages.

Unicode

Unicode is an international standard that aims to give every character in every writing system its own code, including emoji. It uses more bits per character than ASCII, so text can take more storage, but it allows documents to be shared worldwide. Its first 128 codes match ASCII.

T Key terms from the mark schemes

ASCII
American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.
Credited: U4 2019 Q7, 2023 Q1, 2025 Q4

Unicode
Character set covering almost all languages and emoji, using up to 32 bits per character; includes ASCII.
Credited: U4 2019 Q7, 2023 Q1, 2025 Q4

Exam focus

Questions include expanding ASCII (2019 Q7(a), 2023 Q1(a)), the capacity of 7-bit ASCII (2023 Q1(b)), true/false statements (2019 Q7(b), 2023 Q1(d)), and advantages/drawbacks of each (2025 Q4(d)).

Unit 4 2025 Q4	[15]	Binary conversion, addition and overflow, units, Unicode vs ASCII
Unit 4 2023 Q1	[10]	ASCII acronym and capacity; AND/OR truth table; Unicode vs ASCII
Unit 4 2022 Q1(d)–(g)	[4]	Abstraction, decomposition, search efficiency, ASCII
Unit 4 2019 Q7	[5]	ASCII and Unicode

? Check yourself

1. Expand the acronym ASCII. [1]
2. How many characters can be represented in 7-bit ASCII? Explain. [2]
3. The ASCII code for 'A' is 65. What is the code for 'E'? Give it in denary and binary. [2]

Answers: 1. American Standard Code for Information Interchange (1) 2. 128 characters (1) / because 7 bits give 2^7 = 128 different binary patterns (codes 0–127) (1) 3. E is 4 letters after A, so 65 + 4 = 69 (1) / 01000101 (1)

Data types

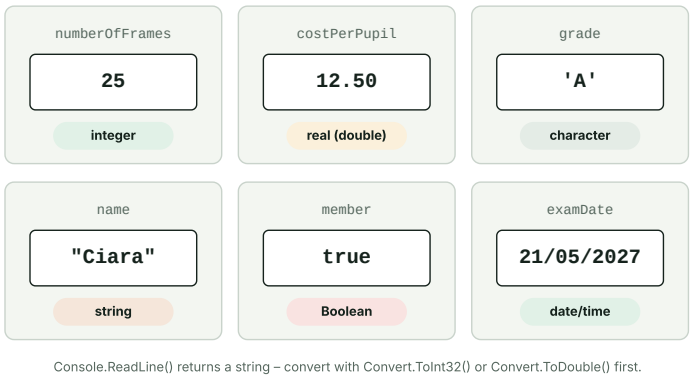
D07
FACT FILE

Specification: students should be able to

- Describe and use appropriately the following data types: numeric (integer and real), date/time, character and string

CCEA fact file: Digital Data 2

Key words: data type, integer, real, double, character, string, Boolean, date/time, casting, type conversion



Six variables, each holding a value of a different data type

Data types

An integer stores a whole number; a real stores a number with a decimal part (double in C#); a character stores one symbol; a string stores text; a Boolean stores one of two values, true or false; date/time stores dates and times. The type controls how much memory is used and which operations make sense. In C#, Console.ReadLine() returns a string, so values must be converted with Convert.ToInt32() or Convert.ToDouble() before arithmetic. Dividing one int by another gives a whole number, so cast to double when a decimal answer is needed.

T Key terms from the mark schemes

Boolean
Data type with only two values, True or False.
Credited: U4 2024 Q8, 2025 Q6

Float / real
Data type for decimal numbers.
Credited: U4 2024 Q3

C C# example

Type conversion (casting)

```
Console.Write("Age? ");
int age = Convert.ToInt32(Console.ReadLine());
Console.WriteLine("Next year you will be " + (age + 1));

double price = Convert.ToDouble("3.99");
Console.WriteLine(price * 2);

int marks = 7;
double average = (double)marks / 2;   // cast: 3.5, not 3
Console.WriteLine(average);
```

Exam focus

Every paper asks for data types in context: 2019 Q1(a) and Q9(a)(ii), 2022 Q2(a) and Q4(b)(ii), 2023 Q3(a), 2024 Q3(b)(iii) and Q8(a), 2025 Q1(a) and Q6(b). Justify by linking to the data, e.g. 'real because the cost includes pence'.

Unit 4 2025 Q6	[25]	Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop
Unit 4 2025 Q3	[15]	Musician hire algorithm; variables and data types; IDE aids; syntax vs logic
Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q8	[18]	String and Boolean types; unit testing; test plan; black vs white box
Unit 4 2024 Q3	[17]	Travel insurance cost algorithm; validation; real data type; WHILE range-check loop

? Check yourself

1. State the most suitable data type for: (a) number of nights booked (b) cost of a booking (c) whether equipment is hired (d) customer name. [4]
2. Describe the Boolean data type. [1]
3. Explain why a race time of 12.84 seconds should be stored as a real. [1]

Answers: 1. (a) integer (1) / (b) real (accept float/double/currency) (1) / (c) Boolean (1) / (d) string (1) 2. A Boolean can hold only one of two values: True or False (1) 3. The time has a decimal (fractional) part, which an integer cannot store, so a real is needed (1)

Boolean operators and truth tables

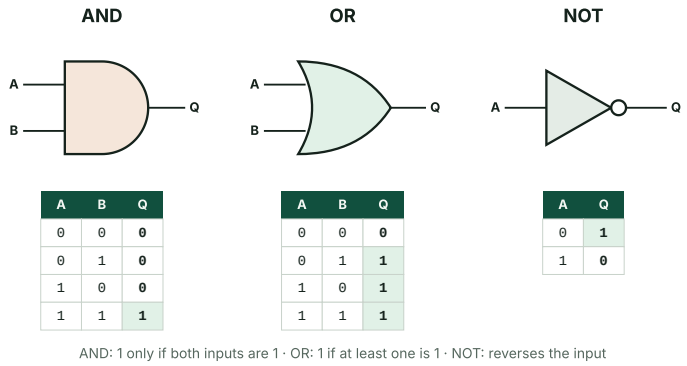
D08 FACT FILE

Specification: students should be able to

- Demonstrate understanding of and use Boolean operators (AND, OR and NOT) and truth tables

CCEA fact file: Digital Data 2

Key words: Boolean, AND, OR, NOT, truth table, logic gate, condition, relational operator



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

A	Q
0	1
1	0

AND, OR and NOT logic gates with their truth tables

Operators

AND gives True only when both inputs are True. OR gives True when at least one input is True. NOT reverses its input. A truth table lists every possible combination of inputs and the output; with n inputs there are 2^n rows. For combined expressions, add an intermediate column for each bracket.

In programs

Conditions in if statements and loops are Boolean expressions. Relational operators (==, !=, <, >, <=, >=) produce true or false, and && (AND), || (OR) and ! (NOT) combine them, e.g. while (choice != "Y" && choice != "N"). A common error is using OR (||) where AND (&&) is needed in a validation loop.

T Key terms from the mark schemes

Truth table
Table showing the output for every combination of inputs.
Credited: U4 2022 Q1, 2024 Q4, 2025 Q1

AND / OR / NOT
AND is true only if both inputs are true; OR is true if at least one is; NOT reverses the input.
Credited: U4 2019 Q8, 2025 Q1

C C# example

Boolean operators in code

```
Console.Write("Age: ");
int age = Convert.ToInt32(Console.ReadLine());
bool hasTicket = true;
if (age >= 12 && age <= 16 && hasTicket)
{
    Console.WriteLine("Welcome to the disco");
}
if (!hasTicket)
{
    Console.WriteLine("Please buy a ticket");
}
```

Exam focus

Truth tables are on every paper: 2019 Q8(e), 2022 Q1(a)–(b), 2023 Q1(c), 2024 Q4(e), 2025 Q1(b) and Q1(d). Fill every row and check intermediate columns.

Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q1	[10]	ASCII acronym and capacity; AND/OR truth table; Unicode vs ASCII
Unit 4 2022 Q1(a)–(c)	[4]	Truth tables for NOT (P AND Q), (P OR Q) AND R; tracing outputs
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Complete the truth table for A AND B for all four input combinations. [2]
2. Complete the truth table for NOT (P AND Q). [2]
3. How many rows does a truth table with three inputs need? Explain. [2]

Answers: 1. A B | A AND B: 0 0 → 0 / 0 1 → 0 / 1 0 → 0 / 1 1 → 1. Award: first three rows all 0 (1) / 1 1 → 1 (1) 2. P Q | P AND Q | NOT (P AND Q): 0 0 → 0 → 1 / 0 1 → 0 → 1 / 1 0 → 0 → 1 / 1 1 → 1 → 0. Award: 1, 1, 1 for the first three rows (1) / 0 for P = 1, Q = 1 (1) 3. 8 rows (1) / because each input can be 0 or 1, so there are $2 \times 2 \times 2 = 2^3 = 8$ combinations (1)

Computational thinking: abstraction and decomposition

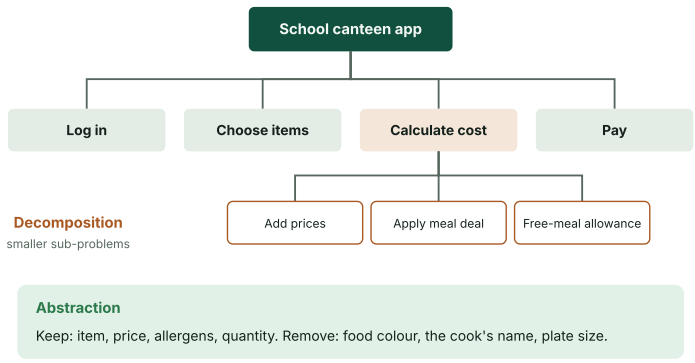
D09 FACT FILE

Specification: students should be able to

- Explain in simple terms the underlying concepts of computational thinking – abstraction and decomposition

CCEA fact file: Digital Design Principles

Key words: computational thinking, abstraction, decomposition, sub-problem, top-down design, structure diagram, algorithm



Decomposition of a school canteen app into sub-problems, with an abstraction note

Abstraction and decomposition

Abstraction means removing details that do not matter so the problem becomes simpler – a model keeps only what is needed. Decomposition means breaking a problem down into smaller sub-problems that are easier to understand, solve and test individually; a structure diagram shows this top-down. Together they let a programmer design an algorithm, a precise sequence of steps that solves the problem.

Benefits

Abstraction reduces complexity and the amount of data and code required. Decomposition makes large problems manageable, lets team members work on separate parts at the same time, allows parts to be reused, and makes testing and debugging easier because each module can be checked separately.

T Key terms from the mark schemes

Top-down design
Breaking the main task into smaller sub-tasks.
Credited: U4 2025 Q7

Decomposition
Breaking a complex problem into smaller, more manageable sub-problems.
Credited: U4 2019 Q6, 2022 Q1, 2023 Q4

Abstraction
Removing details of a problem that are not needed for the solution.
Credited: U4 2019 Q6, 2022 Q1, 2023 Q4

Exam focus

2019 Q6 (QWC), 2022 Q1(d)–(e) (definitions) and 2023 Q4(a) (fill the gaps in a controlled assessment scenario).

Unit 4 2023 Q4	[23]	Decomposition and abstraction; flowchart symbols; syntax/logic errors; functions; menu validation loop
Unit 4 2022 Q1(d)–(g)	[4]	Abstraction, decomposition, search efficiency, ASCII
Unit 4 2019 Q6	[6]	Abstraction and decomposition (levelled)

? Check yourself

1. Define abstraction. [2]
2. Define decomposition. [2]
3. A cinema wants a booking program. Give two details that could be removed through abstraction. [2]

Answers: 1. Removing or hiding details that are not needed to solve the problem (1) / so you focus only on what is important and the problem becomes simpler (1) 2. Breaking a large, complex problem down (1) / into smaller, more manageable sub-problems that can each be solved (and tested) separately (1) 3. Any two relevant details, e.g.: the customer's favourite colour (1) / the colour of the seats/carpet (1) / the plot or actors of the film (1) / the customer's hair colour or clothing (1) / the cinema building's address history (1). Accept any detail not needed for a booking; do not accept film, time, seat, number of tickets or price.

Designing algorithms: flowcharts and pseudocode

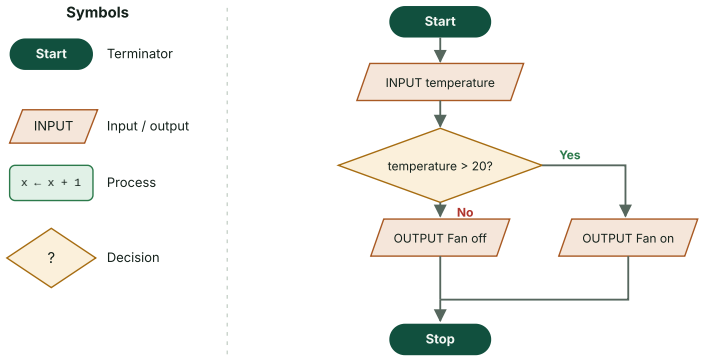
D10 FACT FILE

Specification: students should be able to

- Design solutions using algorithms, flow diagrams and pseudo-code
- Refine a solution to a problem during design

CCEA fact file: Digital Design Principles

Key words: algorithm, flowchart, terminator, process, decision, input/output, pseudocode, sequence, selection, iteration, refinement



Flowchart symbols and a flowchart that turns a fan on above 20 degrees

Representing algorithms

An algorithm is a sequence of precise steps that solves a problem. Flowcharts show it graphically using standard symbols: terminators for start/stop, parallelograms for input/output, rectangles for processes, diamonds for decisions and arrows for flow. Pseudocode describes it in structured English using keywords such as INPUT, OUTPUT, IF, WHILE and FOR, without worrying about a particular language's syntax.

Refinement

Designs are improved iteratively. Dry-running with test values exposes missing steps and logic errors before any code is written; feedback from the user can also add requirements. Refinements might add validation, replace repeated statements with a loop, or split a long algorithm into subroutines.

T Key terms from the mark schemes

Algorithm
A step-by-step plan for solving a problem.
Credited: U4 2019 Q6, 2025 Q3

Flowchart
Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision.
Credited: U4 2022 Q8, 2023 Q4

Pseudocode
Language-independent, English-like instructions describing an algorithm.
Credited: U4 2024 Q7

Exam focus

Writing or completing algorithms is the largest single source of marks on Unit 4 (often 30+ marks). Follow every bullet point in the question, use the variable names given, and make the start and end of each IF and loop clear (2025 examiners). Flowchart questions: 2022 Q8(a), 2023 Q4(b). Defining pseudocode: 2024 Q7(b).

Unit 4 2024 Q7 [35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q4 [23]	Decomposition and abstraction; flowchart symbols; syntax/logic errors; functions; menu validation loop
Unit 4 2022 Q8 [9]	Flowchart selection and iteration; dry runs
Unit 4 2019 Q9 [19]	Test strategies, data types, validation loop, design, discount selection

? Check yourself

1. Define the term algorithm. [2]
2. Describe what is meant by pseudocode. [2]
3. Name and describe the purpose of two flowchart symbols. [4]

Answers: 1. A precise, step-by-step set of instructions (1) / to solve a problem / carry out a task (1) 2. An English-like (structured English) description of an algorithm using programming-style keywords (1) / that is language-independent and ignores exact syntax, so it can be converted into any language (1) 3. Two symbols, each named (1) + purpose (1), e.g.: terminator (rounded rectangle) (1) – start/stop of the algorithm (1) / input/output (parallelogram) (1) – input a value or output a result (1) / process (rectangle) (1) – a calculation or assignment (1) / decision (diamond) (1) – a Yes/No question with two exits (1) / flow line (arrow) (1) – shows the direction of flow (1) / subroutine (double-sided rectangle) (1) – calls a predefined procedure/function (1)

Sorting algorithms: bubble sort and insertion sort

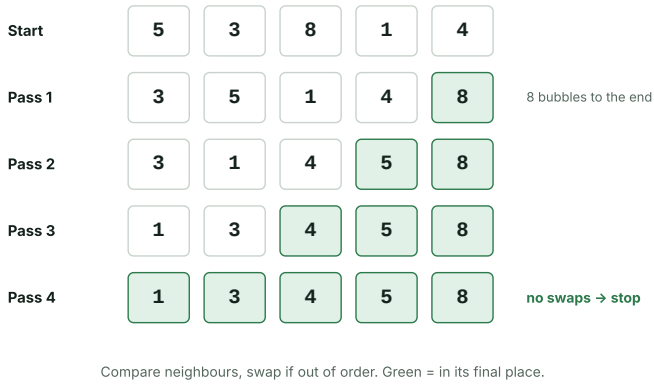
D11 FACT FILE

Specification: students should be able to

- Create and evaluate algorithms, including those for basic sorting
- Simple sorting techniques such as the bubble sort and the insertion sort

CCEA fact file: Programming Constructs 2

Key words: sort, ascending, descending, bubble sort, pass, swap, insertion sort, sorted sub-list, efficiency



Bubble sort passes on 5, 3, 8, 1, 4

Bubble sort

A bubble sort makes repeated passes through the list comparing adjacent pairs and swapping those in the wrong order. After each pass the largest unsorted item reaches its final position at the end. The sort stops when a pass completes with no swaps (or after n − 1 passes). It is easy to write but slow for large lists.

Insertion sort

An insertion sort builds a sorted sub-list at the start. Each new item is taken from the unsorted part and inserted into the correct position in the sorted part, with larger items shifted along. It is generally more efficient than a bubble sort, especially when the data is already nearly in order.

T Key terms from the mark schemes

Bubble sort
Compares adjacent items and swaps them if out of order, repeating passes until no swaps occur.
Credited: U4 2023 Q9, 2024 Q7, 2025 Q1

Insertion sort
Builds a sorted list by inserting each item into its correct position.
Credited: U4 2022 Q5

C C# example

Bubble sort in C#

```
int[] items = { 5, 3, 8, 1, 4 };
bool swapped = true;
while (swapped)
{
    swapped = false;
    for (int i = 0; i < items.Length - 1; i++)
    {
        if (items[i] > items[i + 1])
        {
            int temp = items[i];
            items[i] = items[i + 1];
            items[i + 1] = temp;
            swapped = true;
        }
    }
}
Console.WriteLine(string.Join(", ", items));
```

Exam focus

Recognise how a bubble sort works (2025 Q1(e)), tick true statements and show passes (2024 Q7(d)), order the steps (2023 Q9(c)), show passes on five values (2019 Q12(b)(ii)), and the 2022 QWC on both sorts (Q5). The 2025 report noted pupils need more practical experience of both sorts.

Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q9	[18]	Arrays: update, reset to zero, find maximum; bubble sort steps; searching sorted data
Unit 4 2022 Q5	[6]	Bubble and insertion sort, efficiency (QWC)
Unit 4 2019 Q12	[8]	Linear vs binary search; showing sort passes

? Check yourself

1. Describe how a bubble sort orders a list into ascending order. [4]
2. Show the list after each pass of a bubble sort on 7, 2, 9, 4, 1. [4]
3. Describe how an insertion sort works. [3]

Answers: 1. Any four from: compare the first two items (adjacent pair) (1) / if they are in the wrong order (first larger than second), swap them (1) / move along one position and compare the next pair, repeating to the end of the list – one pass (1) / after each pass the largest remaining item has 'bubbled' to its correct place at the end (1) / repeat passes until a whole pass makes no swaps (or n − 1 passes) (1) 2. Pass 1: 2, 7, 4, 1, 9 (1) / Pass 2: 2, 4, 1, 7, 9 (1) / Pass 3: 2, 1, 4, 7, 9 (1) / Pass 4: 1, 2, 4, 7, 9 (1). (A fifth pass makes no swaps, confirming the list is sorted – not required.) 3. Any three from: the first item is treated as a sorted sub-list (1) / the next item is taken from the unsorted part (1) / and inserted into its correct position in the sorted part (1) / larger items are shifted one place right to make room (1) / repeat until every item has been inserted (1)

Searching algorithms: linear and binary search

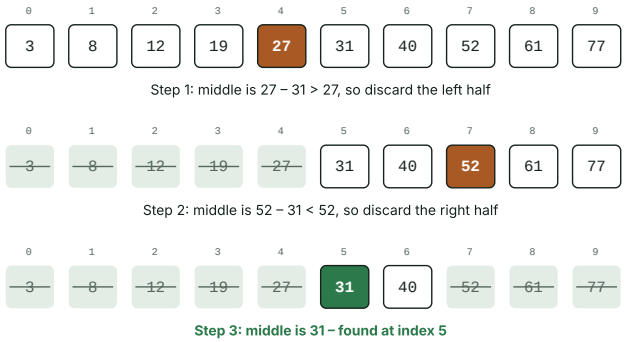
D12 FACT FILE

Specification: students should be able to

- Create and evaluate algorithms, including those for basic searching
- Simple searching techniques such as linear and binary searching

CCEA fact file: Programming Constructs 2

Key words: search, target, linear search, binary search, midpoint, sorted, efficiency, found flag



Binary search for 31 in a sorted list, halving the search each step

Linear search

A linear search checks each item in turn from the start until the target is found or every item has been checked. It works on any list, sorted or not, but on average it must look at half the list and in the worst case all of it, so it is slow for large data sets.

Binary search

A binary search only works on sorted data. It compares the target with the middle item; if they match the search ends, otherwise it discards the half that cannot contain the target and repeats on the other half. Because the list halves each time, it finds items in very few steps even in large lists.

T Key terms from the mark schemes

Linear search
Checks each item in turn until the target is found or the list ends.
Credited: U4 2019 Q12, 2024 Q7

Binary search
Repeatedly checks the middle of a sorted list and discards half; needs sorted data.
Credited: U4 2019 Q12, 2024 Q7

C C# example

Linear search

```
static int LinearSearch(int[] items, int target)
{
    for (int i = 0; i < items.Length; i++)
    {
        if (items[i] == target)
        {
            return i;
        }
    }
    return -1;
}

int[] data = { 12, 7, 31, 4 };
Console.WriteLine(LinearSearch(data, 31));   // 2
```

Exam focus

Efficiency comparison (2022 Q1(f)), definitions and advantages of binary search (2019 Q12), describing both and the sorted-data condition (2024 Q7(e)), choosing the efficient method for sorted data (2023 Q9(c)(iii)).

Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q9	[18]	Arrays: update, reset to zero, find maximum; bubble sort steps; searching sorted data
Unit 4 2022 Q1(d)–(g)	[4]	Abstraction, decomposition, search efficiency, ASCII
Unit 4 2019 Q12	[8]	Linear vs binary search; showing sort passes

? Check yourself

1. Describe how a linear search finds an item. [2]
2. Describe how a binary search finds an item. [3]
3. State the condition that must be met before a binary search can be used. [1]

Answers: 1. Start at the first item and check each item in turn (1) / until the target is found or the end of the list is reached (1) 2. Look at the middle item; if it is the target, it is found (1) / if the target is smaller discard the upper half, if larger discard the lower half (1) / repeat on the remaining half until found or no items are left (1) 3. The data must be sorted (in order) first (1)

Dry runs, trace tables, data requirements and interfaces

D13 FACT FILE

Specification: students should be able to

- Use a dry run to test a solution
- Identify data requirements for a solution and develop an appropriate user interface

CCEA fact file: Digital Design Principles

Key words: dry run, trace table, desk check, data requirements, variable, data dictionary, user interface, prompt

```
a ← 1
WHILE a < 20
    a ← a * 2
END WHILE
OUTPUT a
```

Check the condition every time, including the last one: 32 < 20 is False, so the loop stops and 32 is output – not 16.

a	a < 20 ?	Output
1	True	
2	True	
4	True	
8	True	
16	True	
32	**False**	**32**

A dry run of a WHILE loop recorded in a trace table

Dry runs and trace tables

A dry run means stepping through an algorithm by hand, keeping track of every variable in a trace table and writing down any output. It checks the logic is correct and that the expected output is produced, without needing a computer. Pay particular attention to loop conditions: check the condition every time, including the final time when it becomes false.

Data requirements and interfaces

A designer must identify the data the solution needs – inputs, processed values and outputs – along with sensible names, data types and validation rules. The user interface should give clear prompts and helpful messages, be laid out logically, and suit the intended user.

T Key terms from the mark schemes

Dry run
Working through code by hand, line by line, recording variable values.
Credited: U4 2019 Q10, 2022 Q8, 2025 Q5

Desk checking
Reviewing code by hand to find errors before running it.
Credited: U4 2025 Q5

Trace table
Table recording the values of variables at each step of a dry run.
Credited: U4 2024 Q7, 2025 Q5

C C# example

Predicting loop output

```
for (int i = 5; i < 8; i++)
{
    Console.WriteLine(i * 3);
}
```

Exam focus

Trace and predict questions appear on every paper: 2019 Q10(a) table, 2022 Q8(b) dry runs, 2023 Q2, 2024 Q1–Q2, 2025 Q2 and Q5 (dry run explanation, trace table, final value).

Unit 4 2025 Q5	[7]	Desk checking, trace table, final value
Unit 4 2025 Q2	[12]	Loop output, nested loops, loop count, infinite loop, fixing a condition
Unit 4 2024 Q2	[3]	Tracing FOR and WHILE loops
Unit 4 2024 Q1	[4]	Dry-run a nested IF (shape calculator)
Unit 4 2023 Q2	[4]	Tracing nested selection

? Check yourself

1. Explain how a dry run helps when testing a program. [2]
2. What is a trace table? [1]
3. Predict the output of: for (int i = 1; i < 4; i++) { Console.WriteLine(i * i); } [2]

Answers: 1. Working through the algorithm/program by hand, line by line, without a computer (1) / to find logic errors and check the output is as expected (checking every path) (1) 2. A table that records the value of each variable (and the condition and any output) after each step of an algorithm (1) 3. The loop runs for i = 1, 2, 3 (it stops when i = 4 because i < 4 is false) (1) / output 1, 4, 9, each on a new line, and nothing else (1). No credit for the second mark if 16 is included.

Variables, constants, operators, input, output and assignment

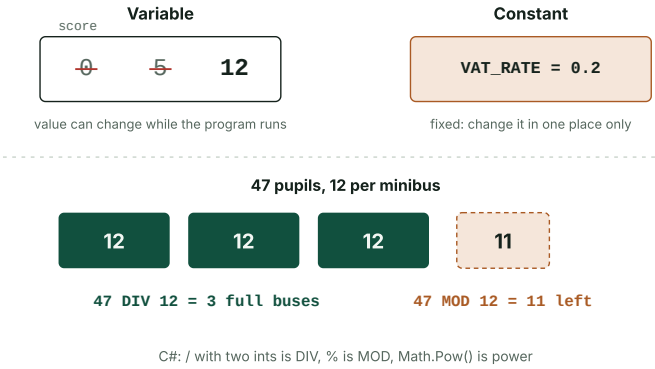
D14 FACT FILE

Specification: students should be able to

- Demonstrate understanding of and use the functionality of variables, constants, and Boolean and arithmetic operators
- Input, output and assignment statements

CCEA fact file: Programming Constructs 1

Key words: variable, constant, identifier, assignment, input, output, arithmetic operator, DIV, MOD, exponent, concatenation



A variable whose value changes compared with a constant, and DIV and MOD

Variables and constants

A variable is a named memory location whose value can change during execution; it should be initialised before use and given a meaningful identifier. A constant is a named value fixed for the whole run. Constants make code easier to read, mean a value only needs changing in one place, and stop the value being altered by accident.

Statements and operators

Input statements read data from the user, output statements display results and assignment stores a value in a variable. Arithmetic operators include +, −, *, /, DIV (whole-number division), MOD (remainder) and powers. Relational operators compare values and Boolean operators combine conditions.

T Key terms from the mark schemes

Variable
A named memory location whose value can change while the program runs.
Credited: U4 2019 Q2, 2025 Q3

Constant
A named value that stays the same while the program runs.
Credited: U4 2019 Q2, 2023 Q3, 2025 Q6

C C# example

Input, output, assignment

```
const double MILES_TO_KM = 1.609;          // constant
Console.Write("Name: ");
string name = Console.ReadLine();          // input
Console.Write("Miles walked: ");
double miles = Convert.ToDouble(Console.ReadLine());
double km = miles * MILES_TO_KM;           // assignment
Console.WriteLine(name + " walked " + Math.Round(km, 2) + " km");   // output
```

Exam focus

Identify variables and constants (2019 Q2(a)), compare them (2019 Q2(b)), define a constant and give a benefit (2023 Q3(b)), two benefits of constants (2025 Q6(c)). Algorithm questions expect correct assignment and output statements using the variable names given.

Unit 4 2025 Q6	[25]	Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop
Unit 4 2023 Q3	[12]	Data types; constants and a benefit; sales total and average algorithm
Unit 4 2019 Q2	[13]	Variables and constants; miles-to-km algorithm with running total
Unit 4 2019 Q1	[18]	Data types, range and presence checks, extending a program, IDE features, translation

? Check yourself

1. Describe the difference between a variable and a constant. [2]
2. Give two advantages of using a constant in a program. [2]
3. What is the result of 23 MOD 5 and 23 DIV 5? [2]

Answers: 1. A variable is a named memory location whose value can change while the program runs (1) / a constant is a named value that does not change while the program runs (1) 2. Any two from: improves readability – a meaningful name instead of a 'magic number' (1) / easier maintenance – the value only needs changing in one place (1) / prevents the value being changed by mistake while the program runs (1). Do not credit a definition of a constant. 3. 23 MOD 5 = 3 (1) / 23 DIV 5 = 4 (1)

Selection: IF, ELSE IF and nested IF

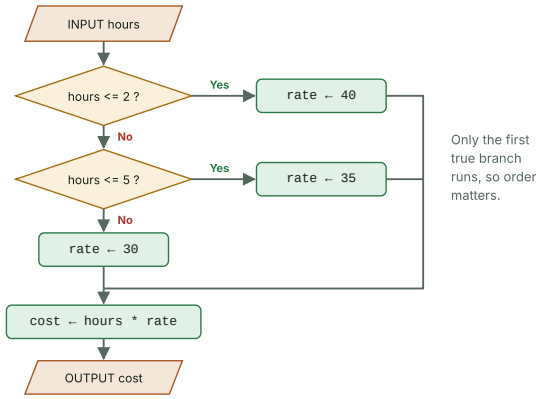
D15 FACT FILE

Specification: students should be able to

- Controlling the flow of a program through sequence, selection and iteration (selection)

CCEA fact file: Programming Constructs 3

Key words: sequence, selection, condition, IF, ELSE, ELSE IF / else if, nested IF, branch



Flowchart of an IF, ELSE IF, ELSE price calculation

Selection

Selection lets a program take different paths. An IF statement runs a block only when its condition is True; IF–ELSE gives an alternative path; ELSE IF (else if in C#) adds further alternatives that are checked in order, so only the first matching branch runs. A nested IF places one decision inside another. Conditions use relational and Boolean operators.

T Key terms from the mark schemes

Sequence
Statements executed one after another in order.
Credited: U4 2019 Q4

Selection
A condition decides which statements run (IF statements).
Credited: U4 2019 Q4, 2025 Q1

C C# example

IF and IF–ELSE

```
Console.Write("Mark: ");
int mark = Convert.ToInt32(Console.ReadLine());
if (mark >= 50)
{
    Console.WriteLine("Pass");
}
else
{
    Console.WriteLine("Fail");
}
```

Exam focus

Read every bullet point – they map to the mark scheme. Use the variable names supplied, use only one condition per IF if told to, include the calculation (e.g. total ← nights * hireCost using variable names, not numbers) and the output statement. Show where each IF ends.

Unit 4 2025 Q6	[25]	Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop
Unit 4 2025 Q3	[15]	Musician hire algorithm; variables and data types; IDE aids; syntax vs logic
Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q3	[17]	Travel insurance cost algorithm; validation; real data type; WHILE range-check loop
Unit 4 2024 Q1	[4]	Dry-run a nested IF (shape calculator)

? Check yourself

1. Describe the difference between sequence and selection. [2]
2. What is output when x = 7? IF x > 10 THEN OUTPUT 'A' ELSE IF x > 5 THEN OUTPUT 'B' ELSE OUTPUT 'C' END IF [1]
3. Write pseudocode that inputs a number of tickets and outputs the total: fewer than 5 tickets cost £8 each, 5 or more cost £6 each. [4]

Answers: 1. Sequence: statements run one after another in order (1) / selection: the program chooses which statements to run depending on whether a condition is True (1) 2. B (1) 3. INPUT tickets
IF tickets < 5 THEN
  price ← 8
ELSE
  price ← 6
END IF
total ← tickets * price
OUTPUT total
Marks: input number of tickets (1) / IF tickets < 5 gives £8 (1) / ELSE (or IF tickets >= 5) gives £6 (1) / total calculated with variables and output (1)

Iteration: FOR, WHILE and REPEAT loops

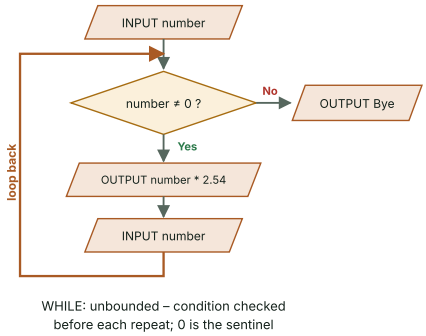
D16 FACT FILE

Specification: students should be able to

- Controlling the flow of a program through sequence, selection and iteration (iteration)

CCEA fact file: Programming Constructs 3

Key words: iteration, loop, FOR, WHILE, REPEAT UNTIL, bounded, unbounded, count-controlled, condition-controlled, running total, counter, sentinel, infinite loop



WHILE: unbounded – condition checked before each repeat; 0 is the sentinel

```
for (int i = 0; i < 5; i++)
```

0 1 2 3 4

FOR: bounded – repeats a set number of times. i < 5 gives 0 to 4, not 5.

do ... while (C#) and REPEAT...UNTIL check at the end: always run once.

Flowchart of a WHILE loop with a sentinel value, and a FOR loop counting

Types of iteration

Bounded iteration repeats a fixed number of times using a FOR loop. Unbounded iteration repeats while (or until) a condition holds: a WHILE loop tests its condition before each repetition, so it can run zero times; REPEAT...UNTIL tests at the end, so it always runs at least once. A sentinel value is a special input that signals the end of data, such as 0.

Common patterns

Initialise a running total or counter before the loop, update it inside, and use or output it after the loop ends. Make sure the loop variable changes inside a WHILE loop so the condition eventually becomes False; otherwise the result is an infinite loop.

T Key terms from the mark schemes

Iteration
Repeating a block of code (loops).
Credited: U4 2019 Q4, 2025 Q2

Infinite loop
A loop whose condition never becomes false, so it never ends.
Credited: U4 2025 Q2

C C# example

FOR loops

```
int total = 0;
for (int count = 1; count <= 5; count++)
{
    Console.Write("Mark: ");
    int mark = Convert.ToInt32(Console.ReadLine());
    total = total + mark;
}
Console.WriteLine("Average: " + (double)total / 5);
```

Exam focus

Loop output prediction (2024 Q2, 2025 Q2), fixing a faulty condition (2025 Q2(c)), running totals (2019 Q2(c), 2023 Q3(c)), WHILE loops with arrays (2024 Q7(c)(iii), 2025 Q8(b)(iii)). Examiners report pupils prefer FOR loops – practise WHILE loops too.

Unit 4 2025 Q8	[20]	Array properties, initialise, count, WHILE totals
Unit 4 2025 Q2	[12]	Loop output, nested loops, loop count, infinite loop, fixing a condition
Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2024 Q2	[3]	Tracing FOR and WHILE loops
Unit 4 2023 Q3	[12]	Data types; constants and a benefit; sales total and average algorithm

? Check yourself

1. Describe the difference between a FOR loop and a WHILE loop. [2]
2. What is output? total ← 0; FOR i ← 1 TO 4: total ← total + i * 2; NEXT i; OUTPUT total [2]
3. Explain what an infinite loop is and give one cause. [2]

Answers: 1. A FOR loop is bounded/count-controlled: it repeats a known/fixed number of times (1) / a WHILE loop is unbounded/condition-controlled: it repeats while a condition is True, checked before each repeat (so it may run zero times) (1) 2. Working: 2 + 4 + 6 + 8 (1) / output 20 (1) 3. An infinite loop never ends because its condition never becomes False (1) / cause, e.g. the loop variable is not changed inside the loop, or the condition uses the wrong comparison (1)

One-dimensional arrays

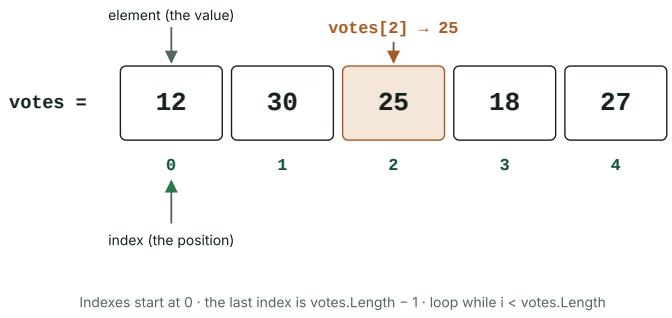
D17 FACT FILE

Specification: students should be able to

- One-dimensional array structures

CCEA fact file: Programming Constructs 1

Key words: array, list, element, index, zero-indexed, initialise, traverse, length



An array of five votes showing elements and zero-based indexes

Arrays

A one-dimensional array stores a collection of values of the same data type under a single identifier. Each element is accessed by its index, which starts at 0. Arrays allow loops to process many values with the same few lines of code. Elements can be read, changed and processed; the index identifies the position, not the value.

Processing arrays

Use a FOR loop over the indexes (for (int i = 0; i < arr.Length; i++)) or a WHILE loop with a counter that increases each time. Typical tasks: set all elements to zero, total all elements, count elements that meet a condition, and find the minimum or maximum by keeping a 'best so far' value.

T Key terms from the mark schemes

Array
A data structure holding values of one data type under one name, accessed by index.
Credited: U4 2024 Q7, 2025 Q8

Index
The position number of an element in an array (starting at 0).
Credited: U4 2025 Q8

C C# example

Creating and accessing

```
int[] votes = { 12, 30, 25, 18, 27 };   // 5 elements, indexes 0-4
Console.WriteLine(votes[0]);      // 12 — first element
Console.WriteLine(votes[4]);      // 27 — last element
votes[2] = 26;                    // update the third element
Console.WriteLine(votes.Length);  // 5

string[] names = new string[3];   // empty array of 3 strings
names[0] = "Ali";
```

Exam focus

Array questions are worth 15–25 marks: 2022 Q7, 2023 Q9, 2024 Q7, 2025 Q8. Common errors: not using a loop when asked, not using the array name given, using the wrong index, placing outputs inside the loop. Know that arrays are zero-indexed and that their contents can be changed (2025 Q8(a)).

Unit 4 2025 Q8	[20]	Array properties, initialise, count, WHILE totals
Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q9	[18]	Arrays: update, reset to zero, find maximum; bubble sort steps; searching sorted data
Unit 4 2022 Q7	[18]	Array indexing; weekly totals and average algorithm; functions and their benefits

? Check yourself

1. Describe what is meant by an array. [2]
2. An array votes holds [14, 22, 9, 31]. What is the value of votes[2]? [1]
3. Write a statement that changes the fifth element of the array heights to 162. [2]

Answers: 1. A data structure that stores many values of the same data type under one name/identifier (1) / each value (element) is accessed by its index, starting at 0 (1) 2. 9 (1) 3. heights[4] ← 162 (C#: heights[4] = 162;). Award: index 4 (zero-indexed fifth element) (1) / correct assignment of 162 to heights (1)

String manipulation

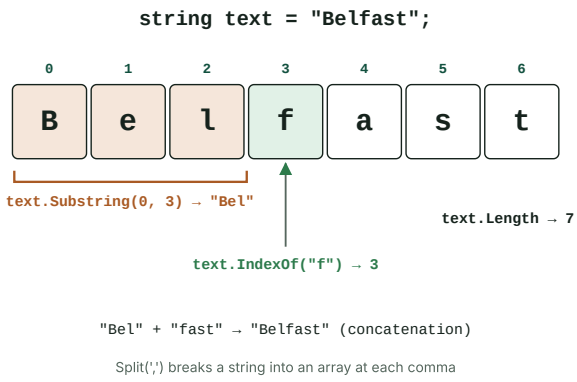
D18 FACT FILE

Specification: students should be able to

- String manipulation functions, including splitting, concatenating, character searching and substring searching

CCEA fact file: Digital Data 2, Programming Constructs 2

Key words: string, concatenation, length, index, substring, Substring, Split, character search, upper, lower



String indexes, Substring, IndexOf and Length for the word Belfast

Key string operations

Concatenation joins strings (with + in C#). The Length property gives the number of characters. Indexing (text[0]) accesses one character and Substring(start, length) extracts part of a string. Searching checks whether a character or substring is present (Contains) or finds its position (IndexOf, -1 if absent). Split(',') breaks a string into an array at a separator such as a comma, which is how CSV records are read.

T Key terms from the mark schemes

Concatenation
Joining strings together.
Credited: U4 2022 Q4

String function
A function that manipulates text, e.g. length, split, find, substring.
Credited: U4 2022 Q4, 2023 Q5

C C# example

String basics

```
string name = "Enniskillen";
Console.WriteLine(name.Length);              // 11
Console.WriteLine(name[0]);                  // E
Console.WriteLine(name[name.Length - 1]);    // n (last character)
```

Exam focus

Name and use string functions: 2022 Q4(c) (join two messages, count characters), 2023 Q5(b) (two functions for password validation). Questions often combine string functions with validation checks.

Unit 4 2023 Q5	[8]	Length, presence and format checks; string functions for a password
Unit 4 2022 Q4	[17]	Tiered pricing algorithm; validation loop 1–50; data types; string functions

? Check yourself

1. What is meant by concatenation? [1]
2. Name a function that returns the number of characters in a string and give an example. [2]
3. word ← 'COMPUTER'. What is the substring from index 3 to index 5 inclusive? [1]

Answers: 1. Joining two or more strings together into one (with + in C#) (1) 2. Length (C#: text.Length) (1) / e.g. "Enniskillen".Length returns 11 (1) 3. "PUT" (C O M P U T E R indexed 0–7: index 3 = P, 4 = U, 5 = T) (1)

Subroutines: user-defined functions and procedures

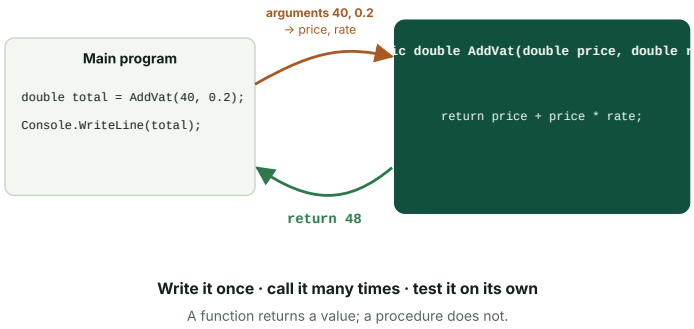
D19 FACT FILE

Specification: students should be able to

- Building reusable code that utilises user-defined functions or methods

CCEA fact file: Programming Constructs 3

Key words: subroutine, procedure, function, method, parameter, argument, return value, local variable, modular, reusable, built-in function



Write it once · call it many times · test it on its own

A function returns a value; a procedure does not.

A main program passing arguments into a function and receiving the return value

Subroutines

A subroutine is a self-contained, named block of code for one task, defined once and called whenever needed. A procedure performs a task; a function also returns a value to the calling code. Data is passed in through parameters. Variables created inside a subroutine are local to it.

Benefits

Subroutines avoid repeating code, make programs shorter and easier to read, and allow each part to be tested and debugged independently. Changes are made in one place. They support decomposition and team working, and well-written subroutines can be reused in other programs.

T Key terms from the mark schemes

Function
A named subroutine that performs a task and returns a value.
Credited: U4 2022 Q7, 2023 Q4

Procedure
A named subroutine that performs a task without returning a value.
Credited: U4 2022 Q7

Parameter
A value passed into a subroutine.

C C# example

Parameters and return values

```
static double AddVat(double price, double rate)
{
    return price + price * rate;
}

double total = AddVat(40, 0.2);
Console.WriteLine(total);      // 48
```

Exam focus

Define a function and give benefits (2022 Q7(b)), two advantages of user-defined functions (2023 Q4(c)(iii)). Unit 5 solutions are expected to use user-defined functions.

Unit 4 2023 Q4 [23]	Decomposition and abstraction; flowchart symbols; syntax/logic errors; functions; menu validation loop
Unit 4 2022 Q7 [18]	Array indexing; weekly totals and average algorithm; functions and their benefits

? Check yourself

1. Define the term function. [2]
2. Describe the difference between a function and a procedure. [2]
3. What is a parameter? [1]

Answers: 1. A named, self-contained block of code that carries out a specific task (1) / and returns a single value to the code that called it (1) 2. A function returns a value to the calling code (1) / a procedure carries out a task but does not return a value (void in C#) (1) 3. A value/variable passed into a subroutine that it needs to carry out its task, listed in its definition (1)

Basic file handling

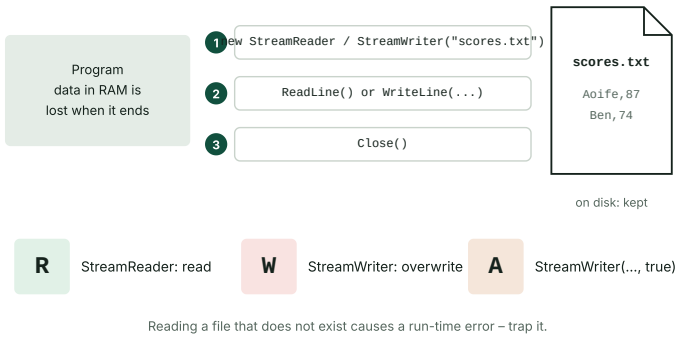
D20 FACT FILE

Specification: students should be able to

- Basic file handling

CCEA fact file: Programming Constructs 3

Key words: file, persistent storage, text file, open, read, write, append, close, file handle, mode, record



Opening, reading or writing, and closing a text file, with read, write and append modes

File handling

Data held in variables is lost when a program ends, so files provide persistent storage. To use a file a program opens it – for reading (StreamReader), writing (StreamWriter, which creates or overwrites) or appending (StreamWriter with true, which adds to the end) – reads or writes the data, then closes it. Text files often hold one record per line with fields separated by commas, which can be separated with Split(',').

T Key terms from the mark schemes

File handling
Opening, reading, writing, appending and closing files to store data permanently.
Credited: U4 2022 Q9, 2023 Q6

C C# example

Writing and appending

```
StreamWriter sw = new StreamWriter("scores.txt");        // write
sw.WriteLine("Aoife,87");
sw.WriteLine("Ben,74");
sw.Close();

StreamWriter sa = new StreamWriter("scores.txt", true);  // append
sa.WriteLine("Cara,91");
sa.Close();
```

Exam focus

2022 Q9(a): two file handling functions besides read and write (e.g. open, close, append, delete). 2023 Q6(b): storing orders for later reports. File handling is required in Unit 5 solutions.

Unit 4 2023 Q6	[10]	Candle shop billing algorithm; storing data in files
Unit 4 2022 Q9	[7]	File handling functions; three error types

? Check yourself

1. Explain why a program might store data in a file. [2]
2. Name and describe two file handling functions other than read and write. [4]
3. Describe the difference between write mode and append mode. [2]

Answers: 1. Data in variables is held in RAM, which is volatile, so it is lost when the program ends (1) / a file stores data permanently (persistent, non-volatile storage) so it can be loaded next time or shared with other programs/users (1) 2. Two functions, each named (1) + described (1), e.g.: open (1) – opens or creates a file ready for reading/writing in a chosen mode (1) / close (1) – closes the file so data is saved and the file is not left locked (1) / append (1) – adds data to the end of a file without deleting existing data (1) / delete (File.Delete) (1) – removes the file (1) / exists (File.Exists) (1) – checks whether a file exists before opening (1) 3. Write mode creates a new file or overwrites (deletes) all existing data in the file (1) / append mode adds new data to the end of the existing file, keeping existing data (1)

Validation, error types and error trapping

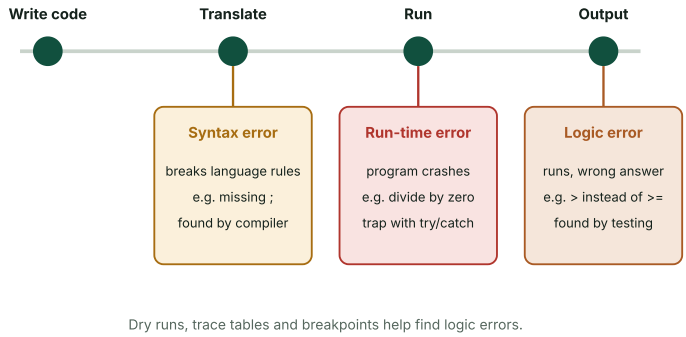
D21 FACT FILE

Specification: students should be able to

- Data validation, including presence, length, type and format checks
- Detection and correction techniques for syntax, execution and logic errors
- Simple error trapping techniques

CCEA fact file: Error Handling

Key words: validation, GIGO, presence check, length check, type check, format check, range check, validation loop, syntax error, execution error, run-time error, logic error, debugging, breakpoint, error trapping, exception, try/catch



When syntax, run-time and logic errors appear during development

Validation

Validation checks that data is reasonable before it is processed: presence (not blank), length (number of characters), type (correct data type), format (correct pattern) and range (within limits). It reduces input errors but cannot prove data is correct. A validation loop repeatedly asks for input while the data is invalid, showing an error message each time.

Errors

Syntax errors break the language's rules and are reported by the translator before the program can run. Execution (run-time) errors happen while the program runs and cause it to crash, e.g. division by zero or opening a missing file. Logic errors let the program run but produce wrong results; they are found by testing against expected outcomes, dry runs, trace tables and debugging tools such as breakpoints.

Error trapping

Error trapping (exception handling) uses try ... catch in C# to catch run-time errors such as FormatException and DivideByZeroException and respond with a helpful message instead of crashing, making the program more robust.

T Key terms from the mark schemes

Syntax error
Breaking the grammar rules of the language; found by the translator before running.
Credited: U4 2019 Q5, 2023 Q4, 2025 Q3

Logic error
The program runs but gives incorrect results because the reasoning is wrong.
Credited: U4 2019 Q5, 2024 Q6, 2025 Q3

Execution / run-time error
An error that occurs while the program runs and causes it to crash, e.g. divide by zero.
Credited: U4 2022 Q9, 2024 Q6

Error trapping
Code that detects a run-time error and handles it instead of crashing, e.g. try/except.
Credited: U4 2022 Q10

C C# example

Validation loops

```
Console.Write("Frames (10-40): ");
int frames = Convert.ToInt32(Console.ReadLine());
while (frames < 10 || frames > 40)
{
    Console.WriteLine("Error: enter a number from 10 to 40");
    Console.Write("Frames (10-40): ");
    frames = Convert.ToInt32(Console.ReadLine());
}
Console.WriteLine("Accepted");
```

Exam focus

Validation: 2019 Q1(b), 2022 Q4(b), 2023 Q4(d) and Q5(a), 2024 Q3(b)–(c), 2025 Q6(d). Errors: 2019 Q5, 2022 Q9(b), 2023 Q4(c), 2024 Q6 (QWC), 2025 Q3(c)(ii). Error trapping: 2022 Q10(b).

Unit 4 2025 Q6	[25]	Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop
Unit 4 2025 Q3	[15]	Musician hire algorithm; variables and data types; IDE aids; syntax vs logic
Unit 4 2024 Q6	[6]	Execution and logic errors and handling (QWC)
Unit 4 2024 Q3	[17]	Travel insurance cost algorithm; validation; real data type; WHILE range-check loop
Unit 4 2023 Q5	[8]	Length, presence and format checks; string functions for a password

? Check yourself

1. Explain why validation cannot guarantee that data is correct. [2]
2. Describe a format check and give an example in a password system. [2]
3. Write pseudocode that repeatedly asks for 'Y' or 'N' until one of them is entered, showing an error message for invalid input. [5]

Answers: 1. Validation only checks that data is sensible/follows set rules (e.g. within a range) (1) / data can be valid but still wrong, e.g. an age of 21 typed instead of 22 passes a range check (1) 2. A format check ensures the data matches a set pattern (1) / e.g. the password must end with a symbol, or contain '@' (1) 3. INPUT answer

```
WHILE answer <> "Y" AND answer <> "N"
   OUTPUT "Error: please enter Y or N"
   INPUT answer
END WHILE
```

Marks: initial input (1) / condition-controlled loop (WHILE or REPEAT...UNTIL) (1) / correct condition using AND (or UNTIL answer = "Y" OR answer = "N") (1) / error message shown only for invalid input, inside the loop (1) / input again inside the loop (1). Accept equivalent C#.

Testing approaches, test plans and test data

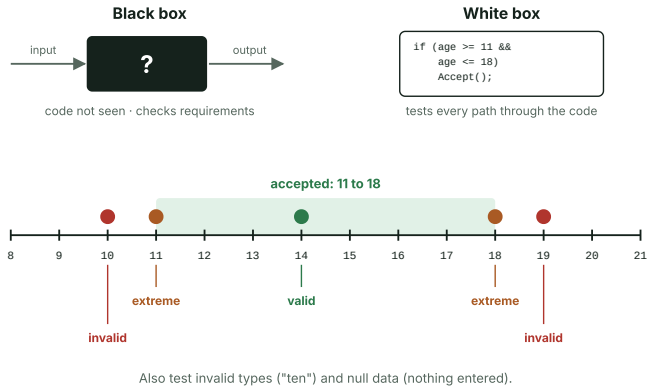
D22 FACT FILE

Specification: students should be able to

- Explain the following iterative approaches to testing: white box; black box; system, unit and integration testing
- Create and use a test plan that identifies test procedures for use during and after development to check a system against success criteria
- Devise and use the following types of test data: valid, invalid and extreme

CCEA fact file: Testing

Key words: iterative testing, white box, black box, unit testing, integration testing, system testing, test plan, test data, valid, invalid, extreme, boundary, null, expected outcome, actual outcome



Black box and white box testing, and test data on a number line for an 11 to 18 age range

Approaches

White box testing examines the internal logic so that every path and condition is executed; it is usually done by the programmer using dry runs and trace tables. Black box testing checks inputs against expected outputs without looking at the code, usually by independent testers or users. Unit testing checks one module on its own; integration testing checks modules work together; system testing checks the whole system against the specification. Testing is iterative – carried out repeatedly during and after development.

Test data and plans

Use valid data (should be accepted), extreme/boundary data (at the limits), invalid data (should be rejected) and null data (nothing entered). A test plan is a table showing each test's number, purpose, test data, type of data, expected outcome, actual outcome and any action taken.

T Key terms from the mark schemes

White box testing
Testing the internal code, paths and logic, usually by the programmer.
Credited: U4 2019 Q10, 2024 Q8, 2025 Q9

Black box testing
Testing inputs and outputs against requirements without seeing the code.
Credited: U4 2019 Q10, 2024 Q8, 2025 Q9

Unit testing
Testing a single module, function or method on its own.
Credited: U4 2022 Q10, 2024 Q8

Integration testing
Testing that separately built units work together.
Credited: U4 2019 Q10, 2023 Q10

System testing
Testing the complete, integrated system against its specification.
Credited: U4 2019 Q10, 2022 Q10

Test plan
Table listing each test, its purpose, test data, expected outcome, actual outcome and action.
Credited: U4 2024 Q8, 2025 Q9

Valid / normal data
Typical data that should be accepted.
Credited: U4 2019 Q9

Extreme / boundary data
Values at the edges of the accepted range.
Credited: U4 2023 Q10, 2024 Q8

Invalid / erroneous data
Data that should be rejected.
Credited: U4 2023 Q10, 2024 Q8

Null data
No data entered.
Credited: U4 2019 Q10, 2024 Q8

Exam focus

2019 Q3 and Q9–Q10, 2022 Q2(c) and Q10(a), 2023 Q10, 2024 Q8, 2025 Q9. Learn clear definitions of each testing type and practise filling test plan rows.

Unit 4 2025 Q9	[11]	Test plan components; black vs white box
Unit 4 2024 Q8	[18]	String and Boolean types; unit testing; test plan; black vs white box
Unit 4 2023 Q10	[9]	Testing types gap fill; invalid and extreme test data
Unit 4 2022 Q10	[9]	Matching five testing methods; error trapping
Unit 4 2022 Q2	[14]	Data type for a race time; lowest of three algorithm; test data

? Check yourself

1. Describe two differences between black box and white box testing. [4]
2. Describe unit testing. [2]
3. Describe the difference between integration testing and system testing. [2]

Answers: 1. Two differences, each stated for both (2 marks each), e.g.: white box testing looks at the internal code, logic and structure (1) – black box testing looks only at inputs and outputs without seeing the code (1) / white box is usually done by the programmer (1) – black box by independent testers or end users (1) / white box aims to test every path, condition and line (e.g. with dry runs/trace tables) (1) – black box checks the functions meet the requirements using a test plan of inputs and expected outputs (1) 2. Testing one module/function/unit on its own (1) / done first, by the developer, before it is combined with other parts (1) 3. Integration testing checks that separately built units work correctly together (the interfaces between them) (1) / system testing checks the complete system against its specification/requirements, using the test plan (1)

Evaluating a solution

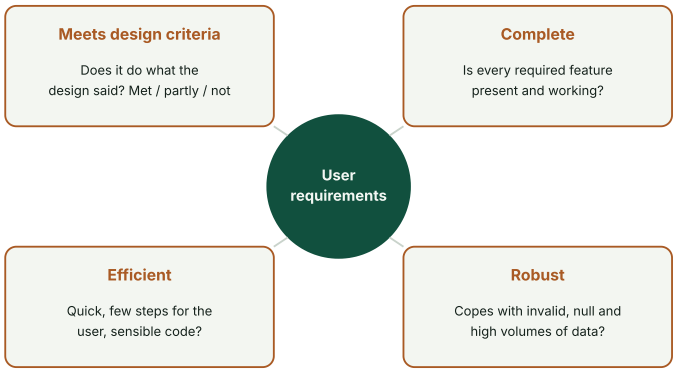
D23 FACT FILE

Specification: students should be able to

- Explain how evaluation is used to ensure that a solution: meets the original design criteria; is a full and complete solution; is an efficient solution; and is a robust solution

CCEA fact file: Testing

Key words: evaluation, user requirements, design criteria, success criteria, complete, efficient, robust, stakeholder, end user, refinement, high-volume testing



Evaluating a solution against the user requirements: meets design criteria, complete, efficient, robust

Evaluation

Evaluation measures a solution against the original user requirements and design criteria. It asks whether the solution meets the design, whether it is complete (every requirement implemented and working), whether it is efficient (fast, economical with resources and easy for the user), and whether it is robust (it copes with invalid, unexpected and large volumes of data without failing). Evidence comes from testing and user feedback. Evaluation should happen throughout development as well as at the end.

T Key terms from the mark schemes

Evaluation
Judging the solution against the user requirements and design.
Credited: U4 2019 Q11, 2024 Q9

User requirements
The agreed list of what the system must do, used to judge the solution.
Credited: U4 2025 Q10

Robustness
The ability to handle invalid, unexpected and high volumes of data without failing.
Credited: U4 2024 Q9, 2025 Q10

Stakeholders
People with an interest in the system: client, end users, developers.
Credited: U4 2019 Q11

Exam focus

2019 Q11 (purpose, stakeholders, requirements, original plan), 2022 Q11, 2023 Q11, 2024 Q9, 2025 Q10. Use the scenario and refer to requirements, testing evidence and robustness.

Unit 4 2025 Q10	[4]	User requirements; robustness
Unit 4 2024 Q9	[7]	Evaluation and robustness
Unit 4 2023 Q11	[7]	Evaluation using requirements and design documents
Unit 4 2022 Q11	[7]	Evaluation statements; complete, robust, efficient
Unit 4 2019 Q11	[10]	Evaluation: purpose, stakeholders, requirements, original plan

? Check yourself

1. Give two reasons why a solution should be evaluated. [2]
2. Explain how user requirements are used during evaluation. [2]
3. What is meant by a robust solution? [2]

Answers: 1. Any two from: to judge how well the solution solves the original problem (1) / to check it meets the user requirements/design criteria (1) / to identify weaknesses, missing features and improvements/refinements (1) / to check it is complete, efficient and robust (1) 2. The user requirements are the agreed standard the solution is measured against – each is checked as a checklist: fully met / partly met / not met (1) / if a feature is missing, the requirements show what must be added before the solution is complete (1) 3. A robust solution handles errors and unexpected input without crashing (1) / it copes with invalid, unexpected/exceptional and high volumes of data (e.g. through validation and error trapping) (1)